# The Collapsed Wing

Knowledge Preconditions over SLAM Maps,
and a Way Known Clear by Elimination

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Three robots enter a building nobody has mapped, and each builds its own map with slam_toolbox from its own laser. A medic must reach a victim in the north wing by one of three ways, any of which rubble may block, and may cross only a way it knows to be clear. Two scouts can each see one doorway; the third way, a passage that winds through collapsed rooms, cannot be seen whole from any spot. Aletheia's policy, of depth 7 over the seven worlds, sends both scouts out at once and has them say what their maps decide; when both doorways prove blocked, the second scout announces the passage clear, which it knows by elimination from what it saw and what it heard. In Gazebo, over 21 seeds, the policy brought the medic to the victim in 21 of 21 runs and never into a blocked way, in 156.1 s on average and 229.1 s at worst. A classical plan assuming every way clear entered 12 blocked ways in 9 runs, with a worst time of 434.4 s; a classical plan that gates the crossing on a scout's check had no plan in the 3 runs where only the passage was clear.

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## 1 Scope

This report accompanies the page *The collapsed wing* of the Epistemic Robotics project and the package `collapsed_wing_demo`. It states the building, the domain and the policy, derives the scout's knowledge by elimination, describes two classical formulations, and reports a batch of 63 runs in Gazebo.

## 2 Environment

The building is 20 by 14 m (Figure 1). The entrance hall, where the medic $m$ starts, lies between a west wing, which the scout $s_2$ enters, and an east wing, which $s_1$ enters. A wall closes off the north wing, where the victim lies, crossed by a west doorway, an east doorway and, between them, a passage of 21 m in three legs through collapsed rooms. The medic's way to the victim is 23 m by either doorway and 31 m by the passage. Rubble, when a way is blocked, stands in a doorway's gap or halfway along the passage's middle leg.

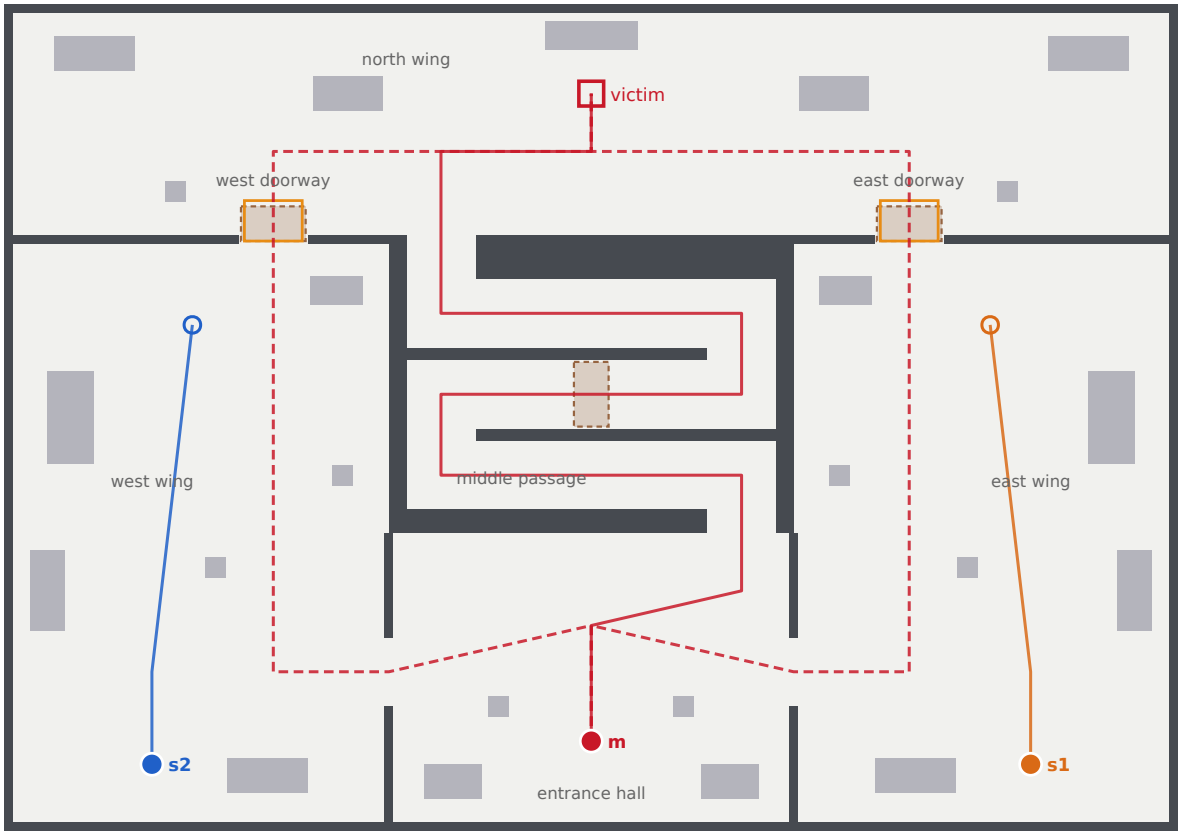



**Figure 1:** The building. Dashed: where rubble stands when a way is blocked. Orange: the region a scout's map must decide. Lines: the scouts' routes and the medic's by each way.

Every robot runs its own slam_toolbox [6] in its own namespace and map frame; no map is shared. A scout's map decides its doorway when its cells in the outlined region say so: blocked when at least 6 are known occupied, clear when none is and at least 90 % of its 10 cm blocks hold a known free cell. The rays of the building are cast in every world before any run: a scout at its vantage decides its own doorway correctly and not the other, and no stopping place has the passage's rubble in sight.

## 3 Domain

The domain has the atoms $\mathit{clear}_w$, $\mathit{clear}_m$ and $\mathit{clear}_e$, the scouts' positions, their roles and $\mathit{with\text{-}victim}(m)$, and four kinds of action, given as event models [1] and planned over as in [2]. A drive is public and ontic. An inspection is semi-private sensing: the scout observes the doorway's state, the others that it looked. A message $\mathsf{tell}_d(s)$ is public sensing with one event per value, each requiring $Kw_s\,\mathit{clear}_d$ beside the fact. A crossing $\mathsf{cross}_d(m)$ is public and ontic, with precondition $K_m\,\mathit{clear}_d$ and effect $\mathit{with\text{-}victim}(m)$. Only $s_1$ can reach the east doorway and only $s_2$ the west; no action observes the passage. The initial model has seven designated worlds, in which $\mathit{clear}_w \vee \mathit{clear}_m \vee \mathit{clear}_e$ is common knowledge and no robot knows more, and the goal is $\mathit{with\text{-}victim}(m)$.

## 4 The policy

Aletheia's AO* [3] returns, at depth 7 after 424 expansions, a policy with three leaves:

```
goto-e(s1) · goto-w(s2) · inspect-e(s1)
   on e-inspect-e-clear:  tell-e(s1) · cross-e(m)
   on e-inspect-e-blocked:  tell-e(s1) · inspect-w(s2)
      on e-inspect-w-clear:  tell-w(s2) · cross-w(m)
      on e-inspect-w-blocked:  tell-m(s2) · cross-m(m)
```

### 4.1 Knowledge by elimination

In the last branch, after $s_2$'s inspection of the west doorway has shown it blocked and $s_1$'s message has said the east doorway blocked, every world $s_2$ considers possible satisfies $\neg\mathit{clear}_w$ and $\neg\mathit{clear}_e$. Each of them also satisfies $\mathit{clear}_w \vee \mathit{clear}_m \vee \mathit{clear}_e$, which is common knowledge from the start and which no event removes, so each satisfies $\mathit{clear}_m$: $K_{s_2}\,\mathit{clear}_m$ holds, and with it $Kw_{s_2}\,\mathit{clear}_m$, the precondition of the clear event of $\mathsf{tell}_m(s_2)$. That message is public, and after it $K_m\,\mathit{clear}_m$ holds, the precondition of $\mathsf{cross}_m(m)$. The passage is crossed without any robot having observed it. The order of the policy follows from this: $s_1$ speaks before $s_2$ looks in the branch where the east doorway is blocked, because $s_2$'s conclusion requires what $s_1$ says.

The two drives come first and, being independent, are dispatched together by the executor, so that both scouts reach their doorways while the first is still to be decided. No branch has the medic cross a way it has not been told is clear.

## 5 Classical planning

A classical planner cannot state that the medic must know the way. In the *optimistic* formulation every way is assumed clear and POPF [4], through PlanSys2 [5], sends the medic by the shortest; when the medic's laser finds rubble on its route the drive fails, the mission removes that way's assumption, places the medic in the way and replans, which backs it out and sends it by another. In the *gated* formulation the crossing of a way requires a fact $\mathit{checked_clear}_d$ that only a scout's check gives, the knowledge precondition written as a fact; a check is planned as if it will find the doorway clear, and the mission replans when it does not. No check of the passage exists, so where it alone is clear the gated formulation has no plan, unless the conclusion by elimination is written into the domain by hand.

## 6 Measurement

A run reaches the victim when the medic ends within 0.5 m of it, by its own odometry. It enters a dead end when the medic sets out on a way and its laser finds rubble. A crossing is told when its way was said clear on the radio before the medic set out. Every change of a scout's doorway

decision is checked against the world, and mission time is measured in simulated seconds from a common start.

## 7 Evaluation over 21 seeds

Table 1 gives the outcome per formulation and Table 2 per world.

| formulation | reached | time (s) | worst (s) | dead ends | told | no plan | medic (m) |
|---|---|---|---|---|---|---|---|
| epistemic policy | 21/21 | 156.1 | 229.1 | 0 in 0 | 21 | 0 | 23.4 |
| classical, optimistic | 21/21 | 188.3 | 434.4 | 12 in 9 | 0 | 0 | 37.5 |
| classical, gated | 18/21 | 156.1 | 180.9 | 0 in 0 | 0 | 3 | 22.4 |

**Table 1:** Runs out of 21 per formulation. Time and medic: means over the runs that reached the victim. Told: crossings of a way said clear on the radio beforehand.

| world | epistemic policy | classical, optimistic | classical, gated |
|---|---|---|---|
| west clear, middle clear, east clear | 3/3, 144 s | 3/3, 109 s | 3/3, 144 s |
| west clear, middle clear, east blocked | 3/3, 144 s | 3/3, 108 s | 3/3, 180 s |
| west clear, middle blocked, east clear | 3/3, 144 s | 3/3, 109 s | 3/3, 144 s |
| west clear, middle blocked, east blocked | 3/3, 144 s | 3/3, 109 s | 3/3, 180 s |
| west blocked, middle clear, east clear | 3/3, 144 s | 3/3, 228 s, 3 dead | 3/3, 144 s |
| west blocked, middle clear, east blocked | 3/3, 227 s | 3/3, 427 s, 6 dead | no plan |
| west blocked, middle blocked, east clear | 3/3, 145 s | 3/3, 229 s, 3 dead | 3/3, 145 s |

**Table 2:** By world: runs that reached the victim, mean time, and blocked ways entered.

**The policy never sent the medic into a blocked way**, and every way it crossed had been said clear on the radio before it set out. Assuming every way clear, the optimistic medic took the west doorway first; where it was clear it reached the victim 35.6 s sooner than the policy on average, having not waited for a scout. Where it was blocked the medic found the rubble with its own laser, backed out and tried the next way, and these 9 runs took 122.5 s longer than the policy's on average. Paired by seed, the policy was faster in 9 of 21 seeds, and its worst time was 229.1 s against 434.4 s.

The gated formulation follows the policy's order but not its schedule. Where the east doorway was clear the two coincide, within 0.5 s on average. Where it was blocked and the west was clear, the gated plan sends s2 only after s1's check has failed, and arrived 35.5 s after the policy, whose s2 had driven out at the start. Where only the middle passage was clear it had no plan, in all 3 runs.

Every robot's own SLAM map held, at the end, 51, 44 and 152 $m^2$ of the building on average for s1, s2 and the medic. No scout's map decided a doorway wrongly in any run.

## 8 Implementation defects

ePlanSys's executor dispatches together consecutive actions of a policy whose vocabulary actions neither write what the other reads. With each vocabulary action touching only its own robot's fact, the medic's crossing was dispatched beside the message it waits for, and the knowledge check stopped it before it moved. The vocabulary now carries a fact for the radio, which every message writes and every crossing reads.

## 9 Limitations

Aletheia's AO* minimises the depth of the worst branch and has no durations: it orders the scouts' looks by depth, so the medic goes by the east doorway whenever it is clear, although the west is as

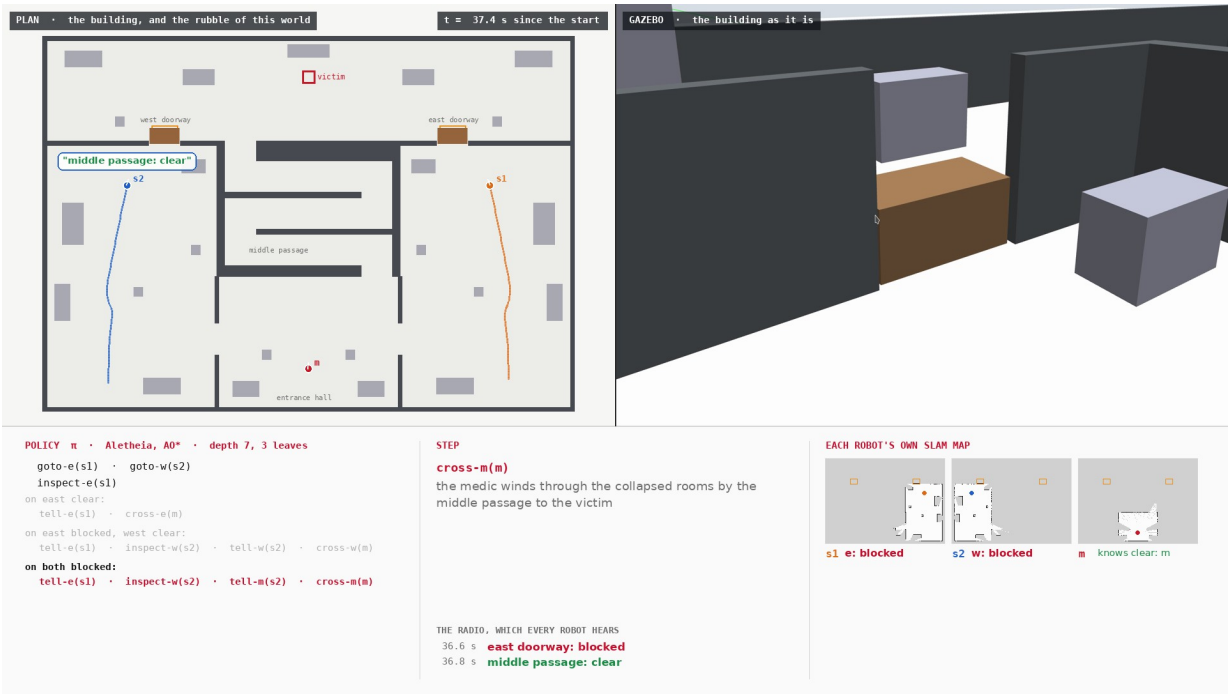



**Figure 2:** The policy on seed 6, where only the middle passage is clear, as $s_2$ announces it clear.

long. The radio reaches every robot and never fails. That at least one way is clear is common knowledge from the start, and is what makes the conclusion by elimination sound; in a building in which every way might be blocked the last branch would have no guarantee.

The drives follow routes on a roadmap, steered by a dynamic window on each robot's own laser, and not a planner over the SLAM map: the maps decide what is known, not where a robot drives. The robots' poses come from the simulator's odometry. Mission time includes the planners' wall time, and the batch ran beside other simulations at a real-time factor near 0.4.

## References

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